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TEYSSEYRE et al.(10) **Pub. No.: US 2025/0279405 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **POWER MODULE PACKAGE FOR DIRECT COOLING MULTIPLE POWER MODULES****Publication Classification**(71) Applicant: **SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC**, Scottsdale, AZ (US)(72) Inventors: **Jerome TEYSSEYRE**, Singapore (SG);
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Jooyang EOM, Gimpo-si (KR)(73) Assignee: **SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC**, Scottsdale, AZ (US)(51) **Int. Cl.****H01L 25/00** (2006.01)
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H01L 25/18 (2023.01)(52) **U.S. Cl.****CPC** **H01L 25/50** (2013.01); **H01L 23/3677** (2013.01); **H01L 23/467** (2013.01); **H01L 23/473** (2013.01); **H01L 25/072** (2013.01); **H01L 25/18** (2013.01)

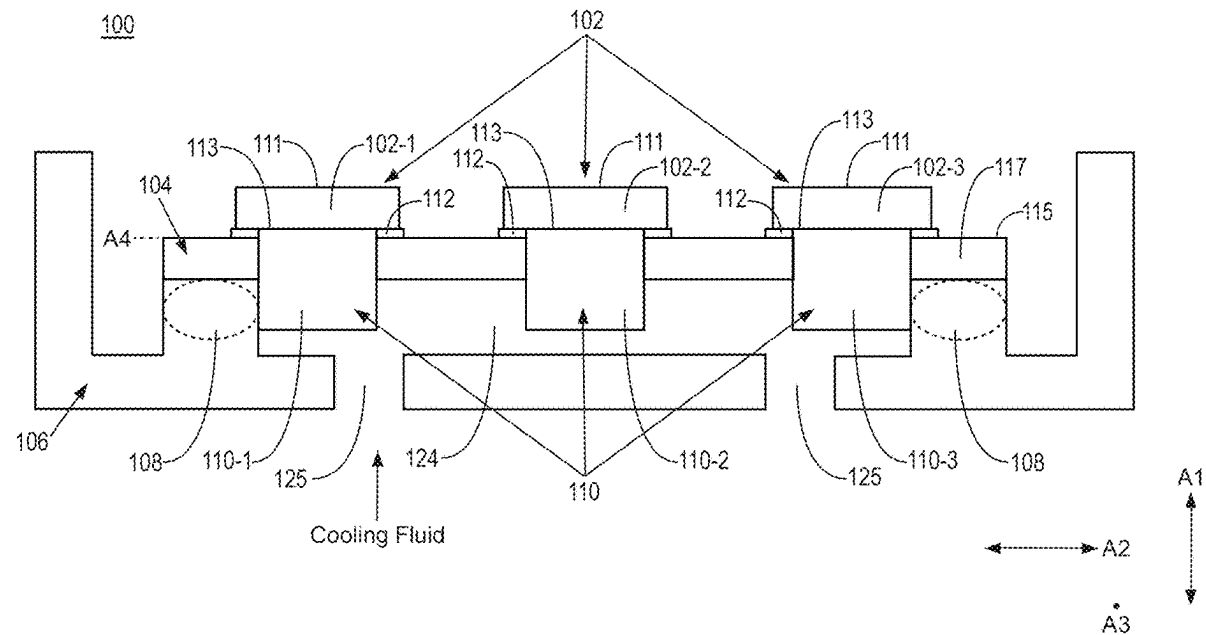
(57)

ABSTRACT

According to an aspect, a power module package includes a plurality of power modules including a first power module and a second power module, a plurality of heat sinks including a first heat sink coupled to the first power module and a second heat sink coupled to the second power module, and a module carrier coupled to the plurality of power modules, where the module carrier includes a first region defining a first heat-sink slot and a second region defining a second heat-sink slot. The first heat sink extends at least partially through the first heat-sink slot and the second heat sink extends at least partially through the second heat-sink slot. The power module package includes a housing coupled to the module carrier and a ring member located between the module carrier and the housing.

(21) Appl. No.: **19/210,838**(22) Filed: **May 16, 2025****Related U.S. Application Data**

(60) Continuation of application No. 18/542,230, filed on Dec. 15, 2023, now Pat. No. 12,308,364, which is a continuation of application No. 17/822,844, filed on Aug. 29, 2022, now Pat. No. 11,848,320, which is a division of application No. 16/949,896, filed on Nov. 19, 2020, now Pat. No. 11,430,777.



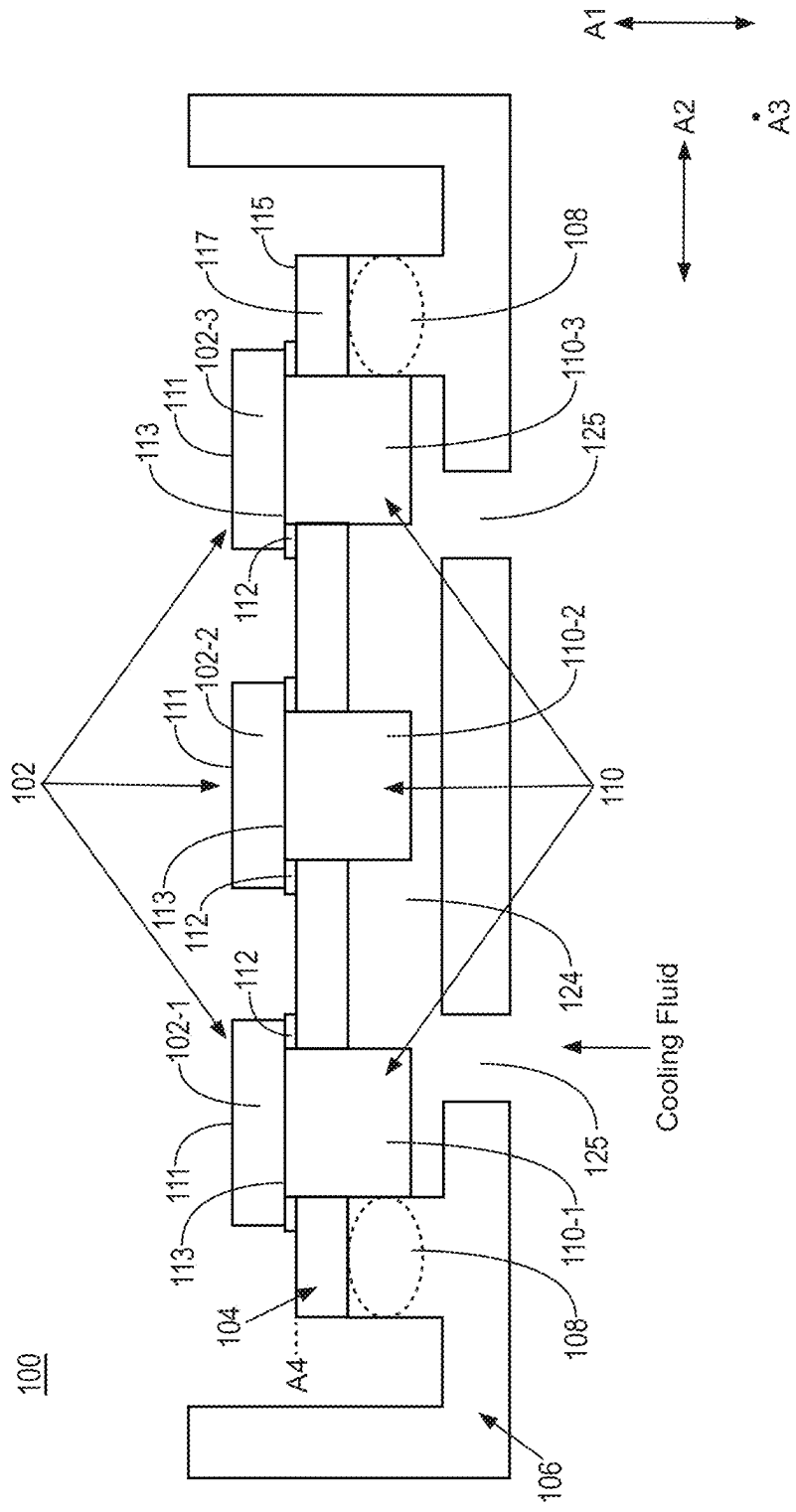


FIG. 1A

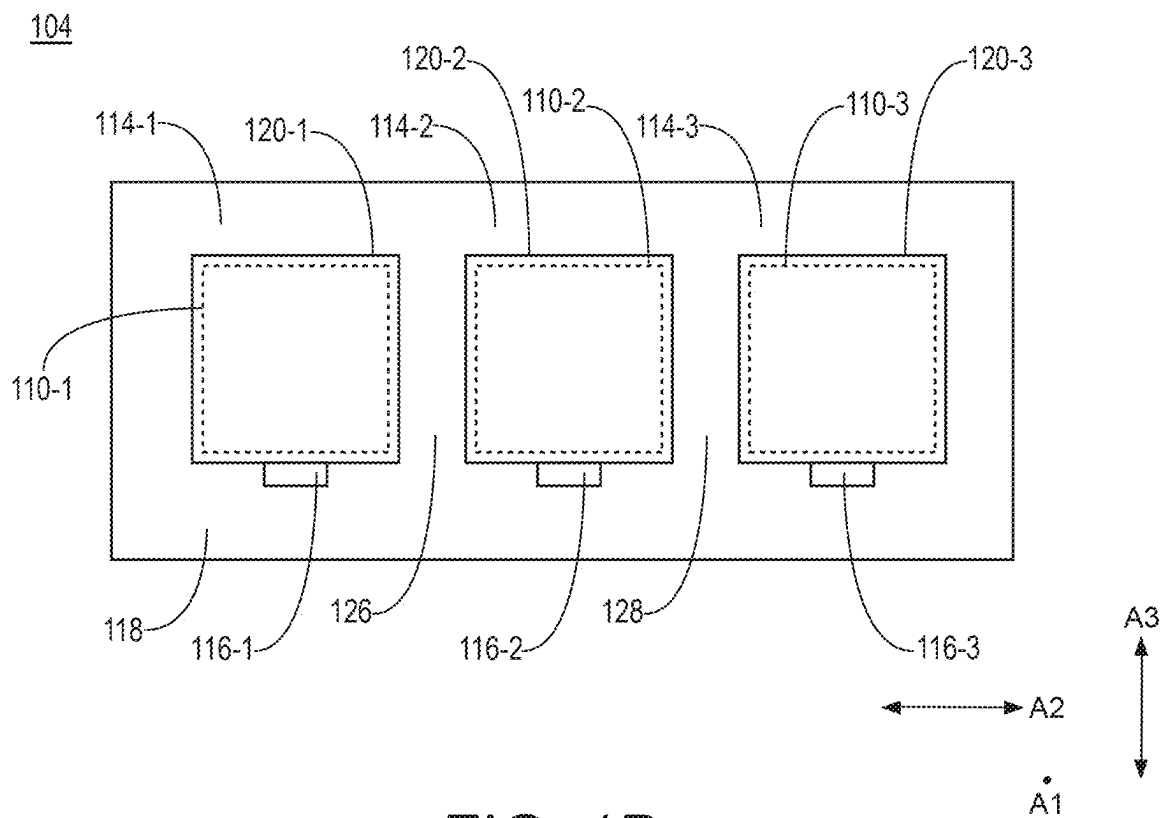


FIG. 1B

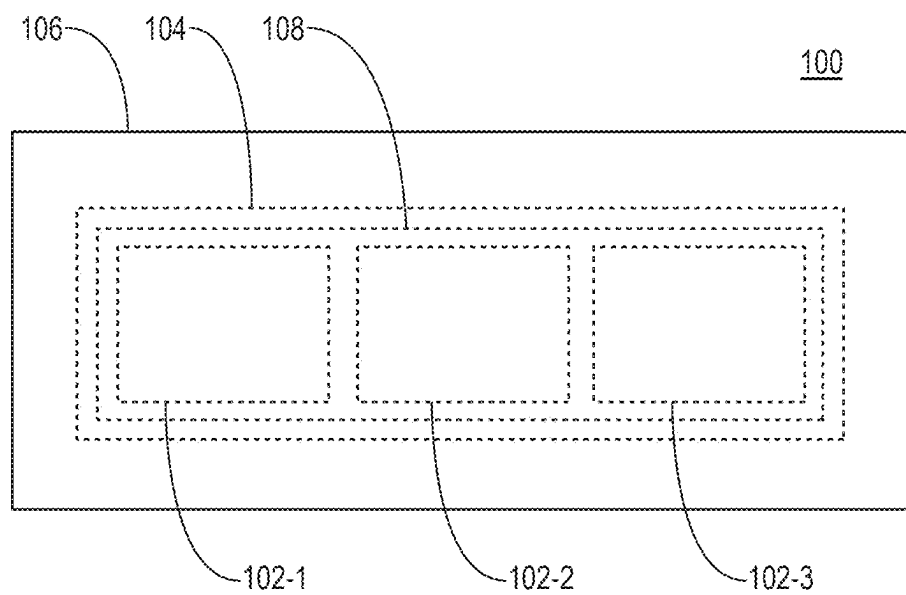


FIG. 1C

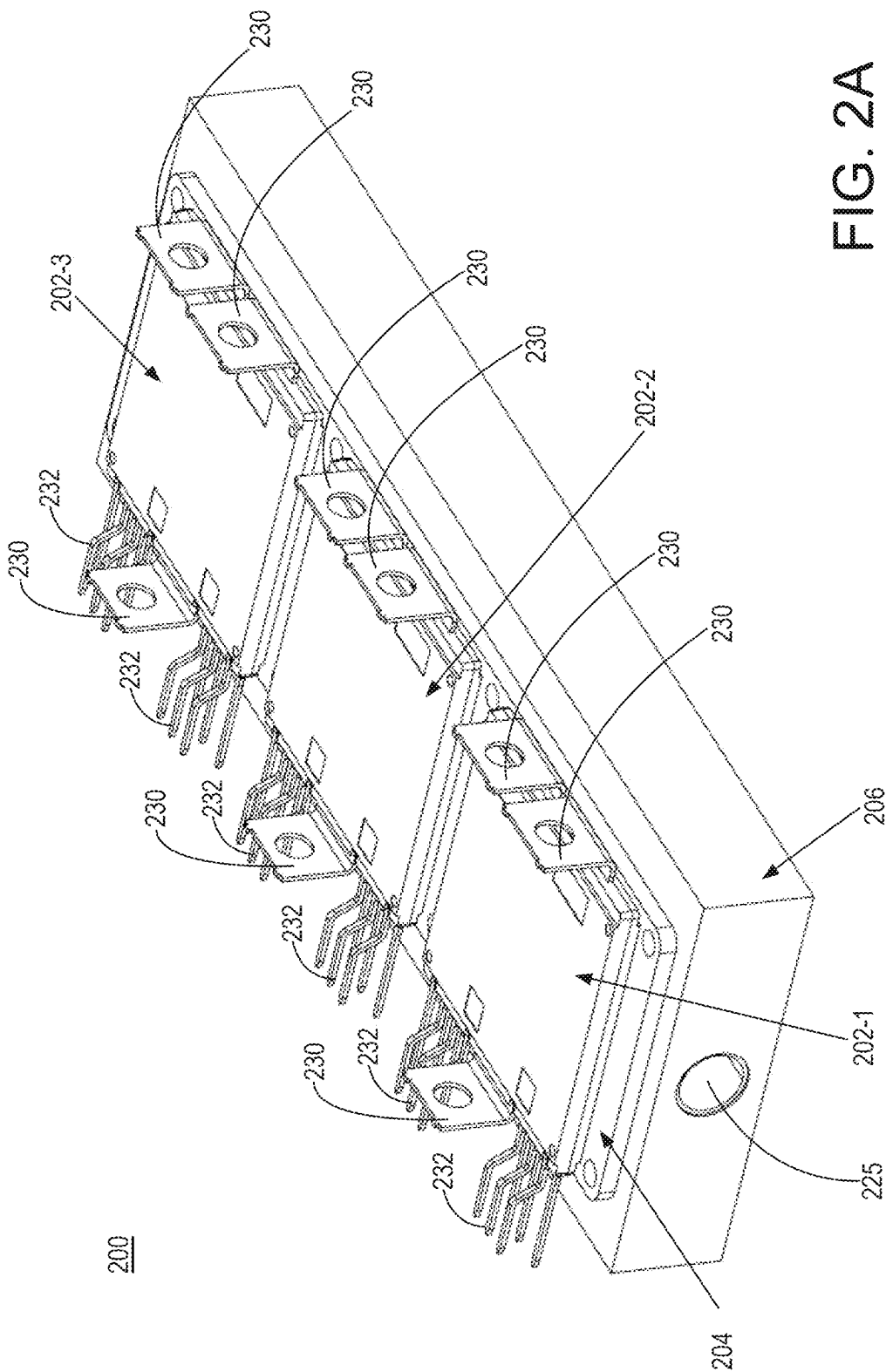


FIG. 2A

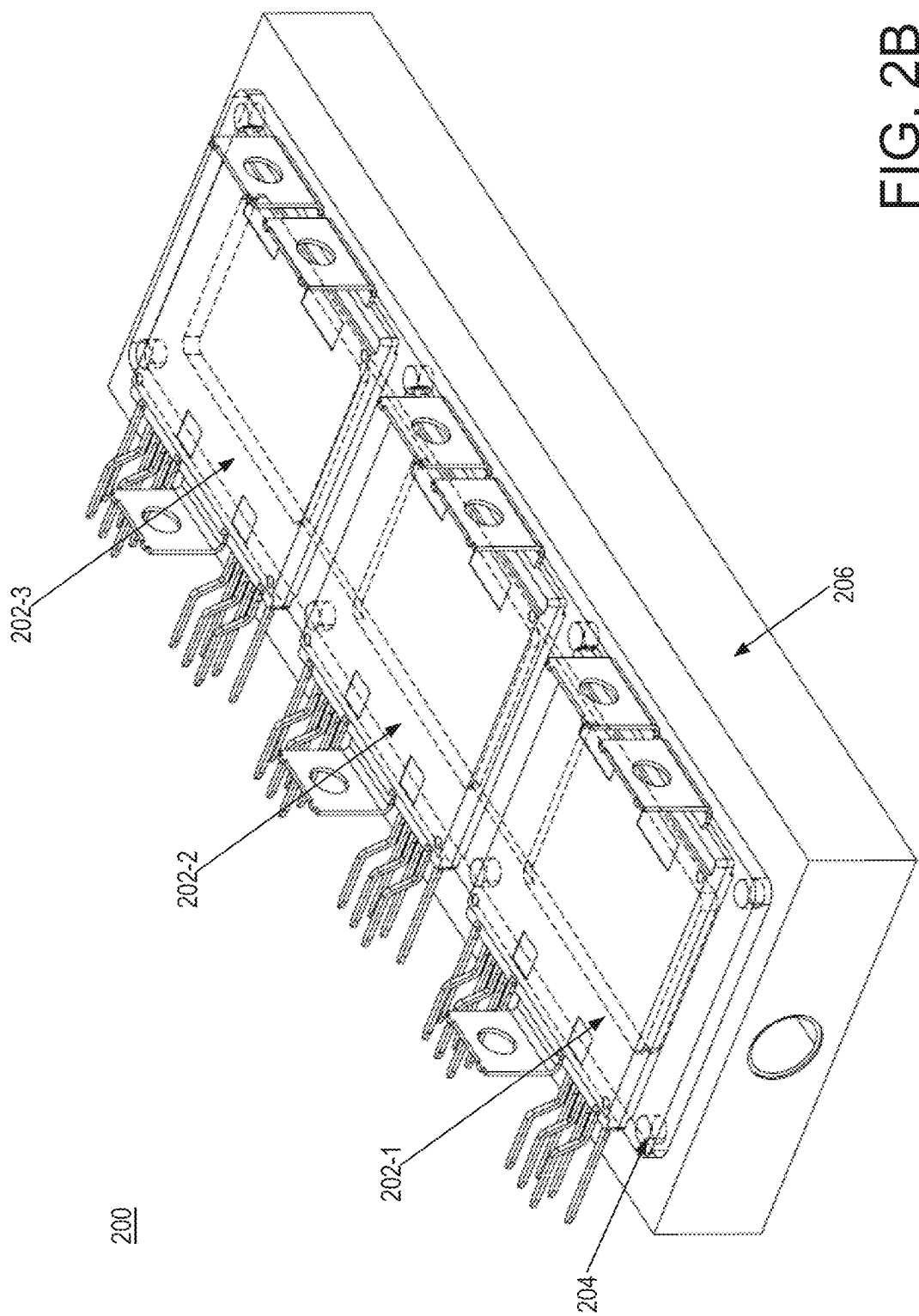


FIG. 2B

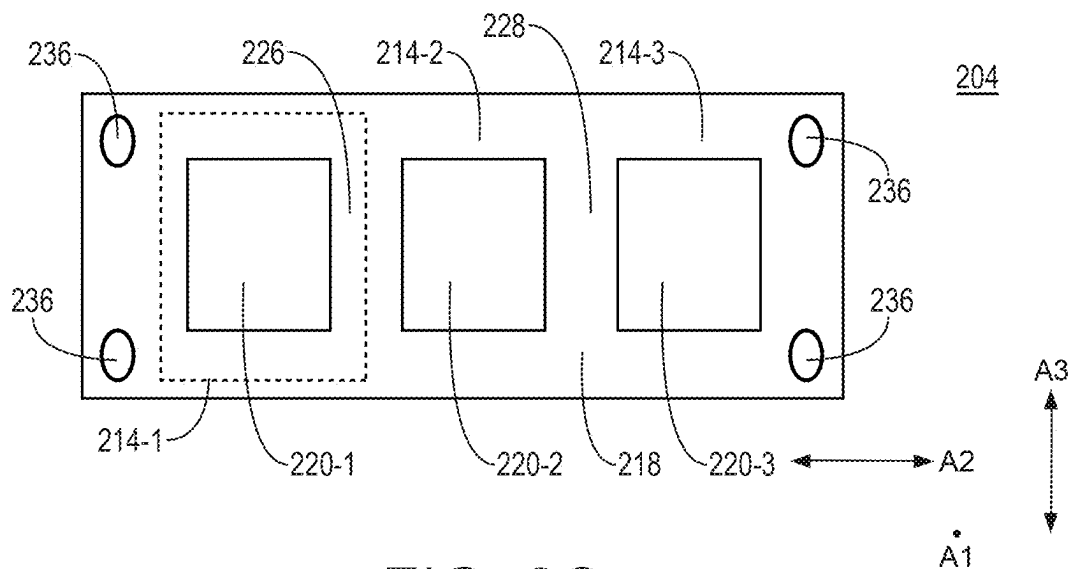


FIG. 2C

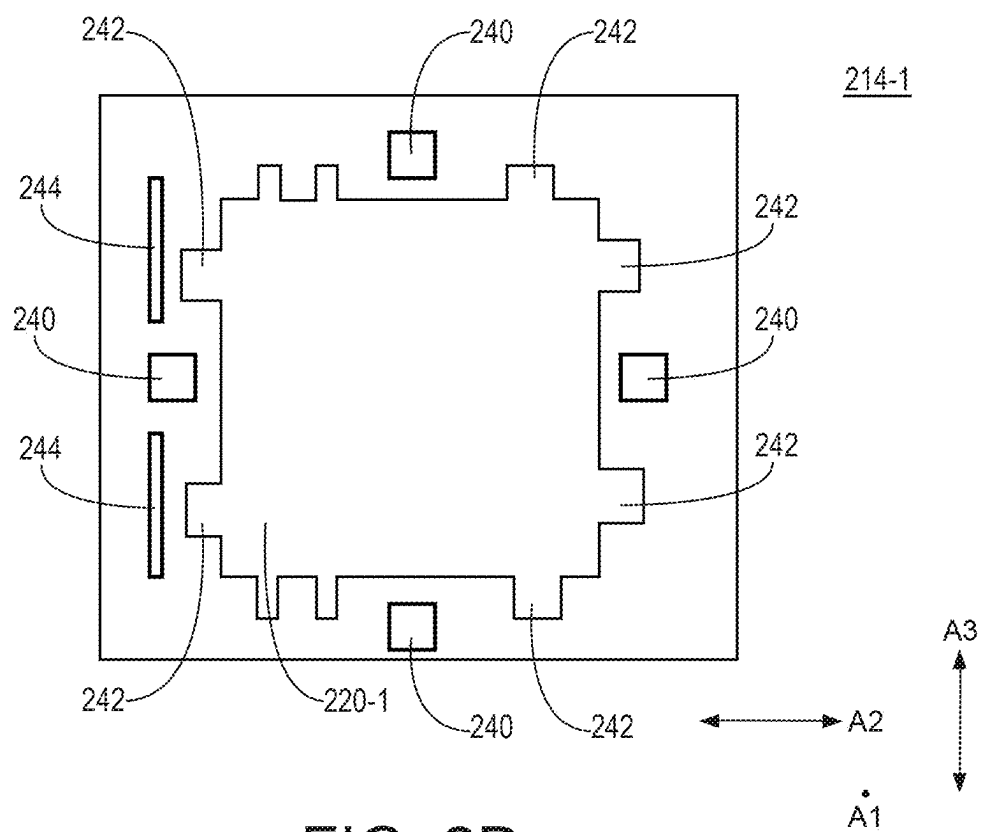


FIG. 2D

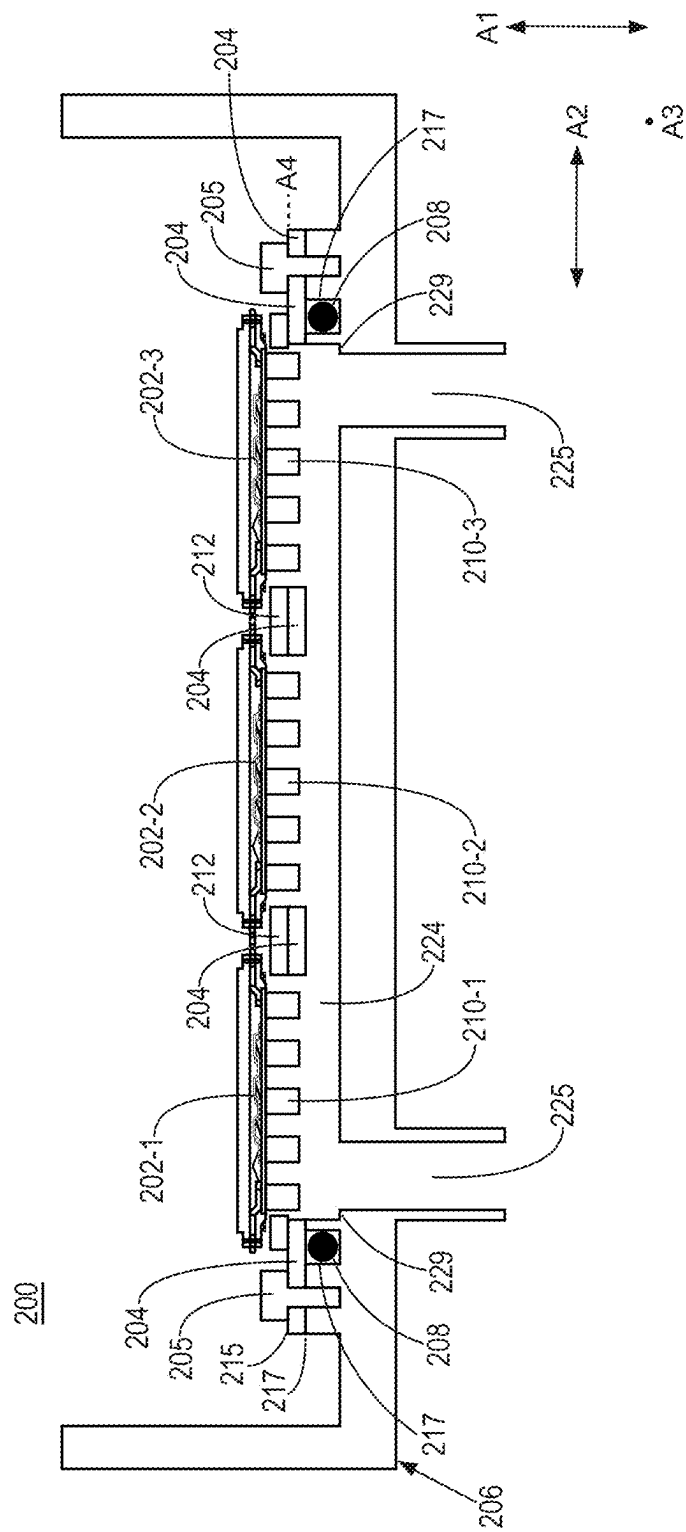


FIG. 2E

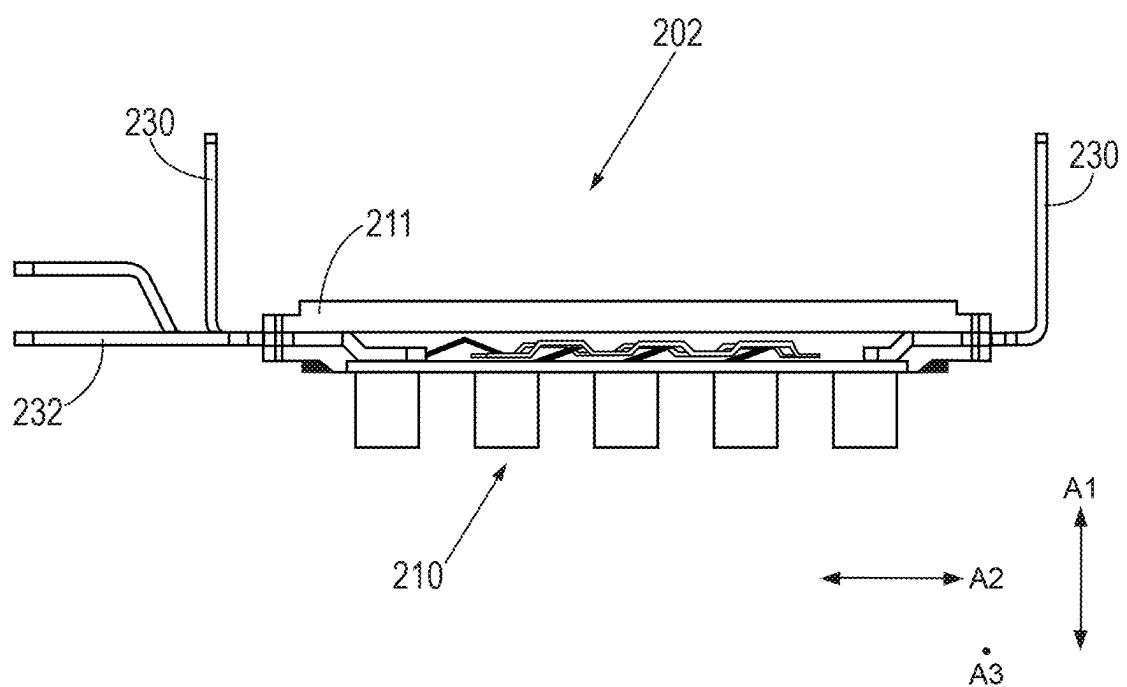


FIG. 2F

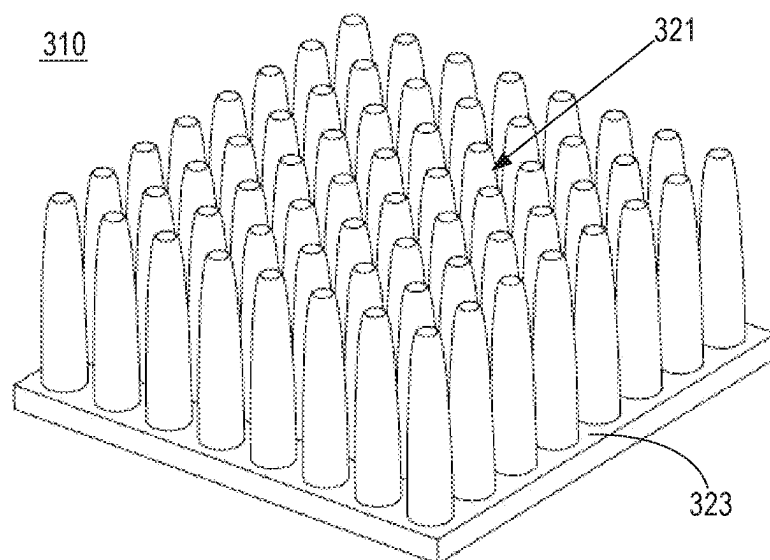


FIG. 3

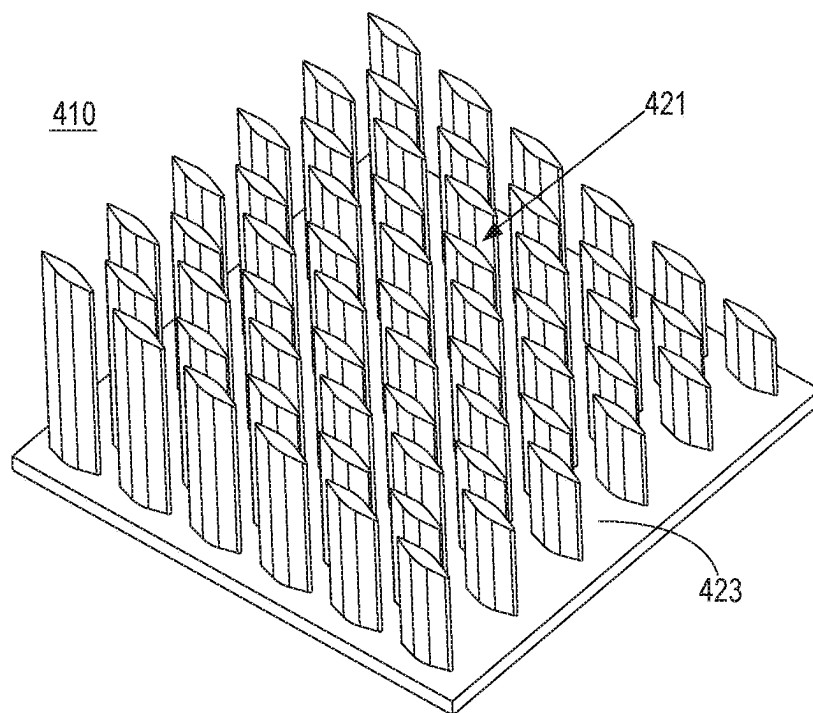


FIG. 4

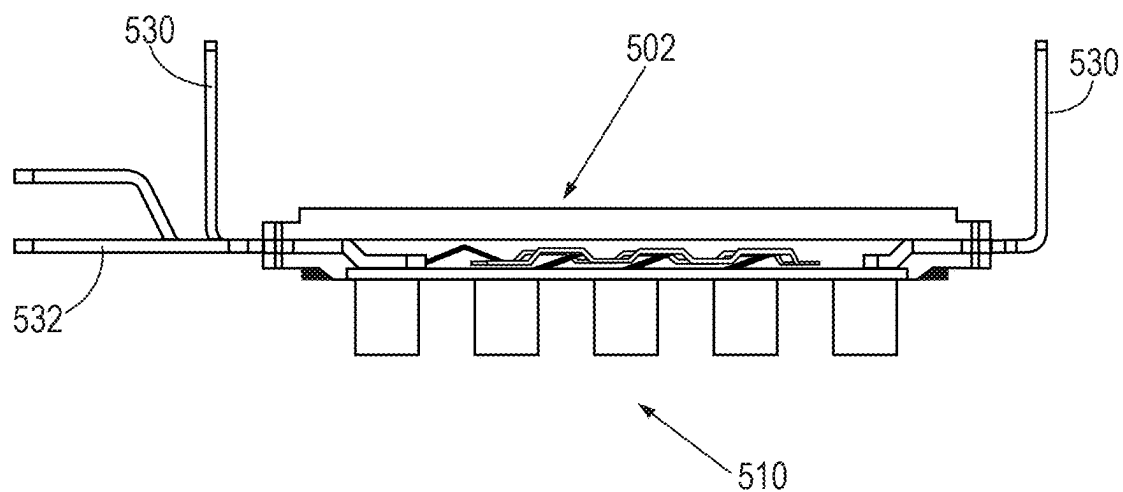


FIG. 5A

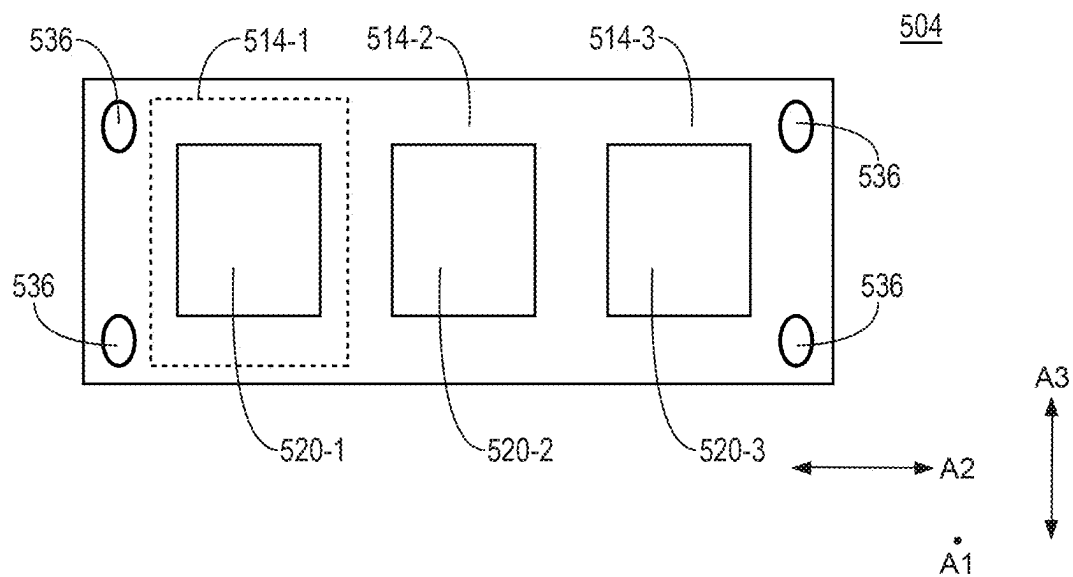


FIG. 5B

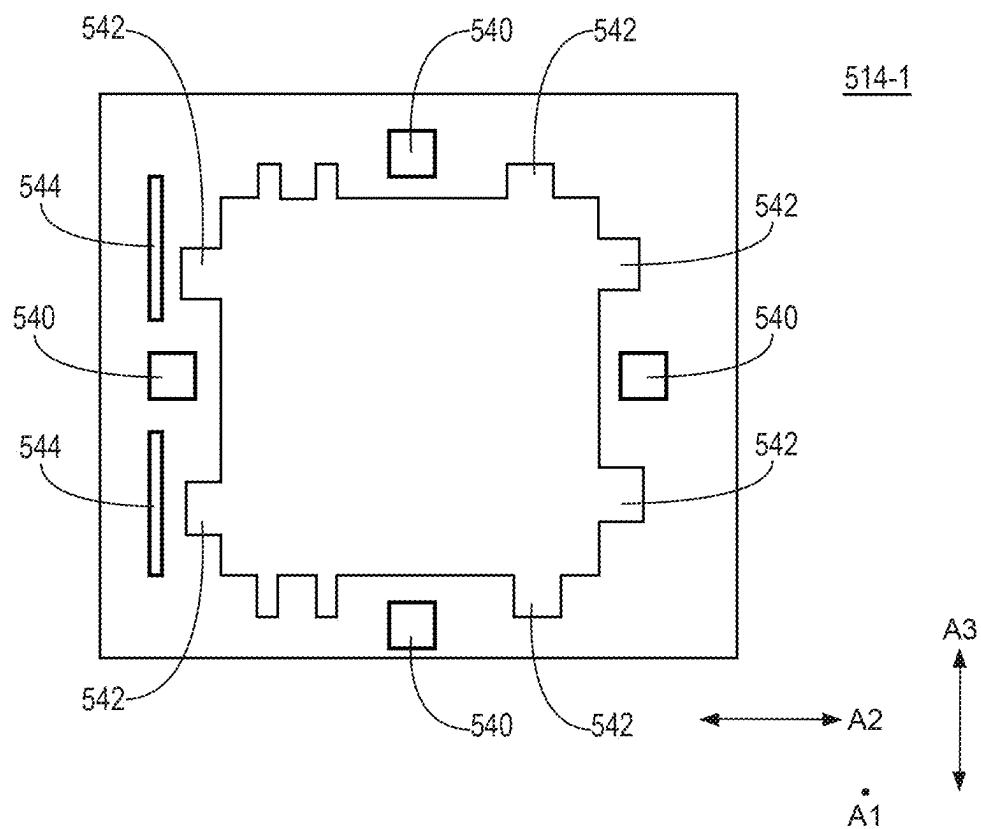


FIG. 5C

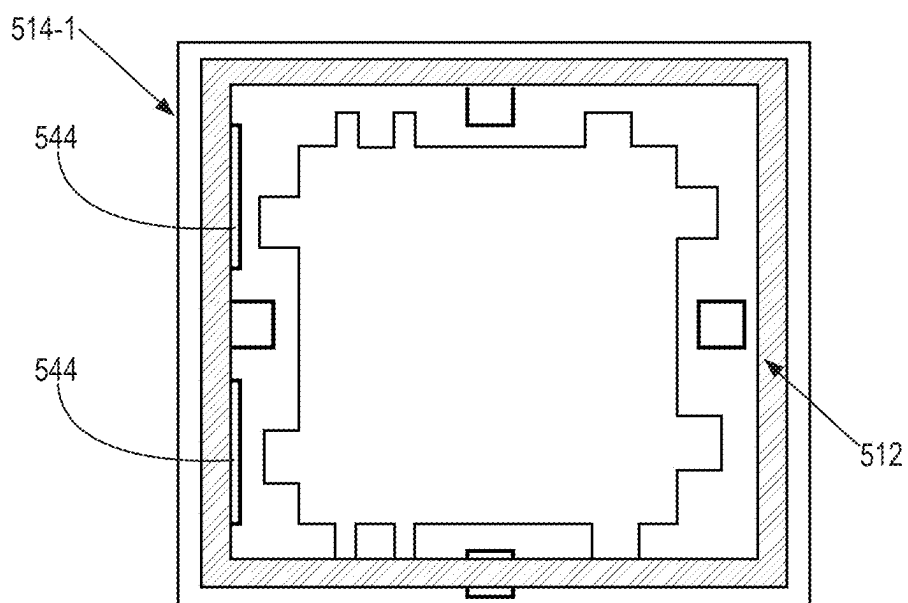


FIG. 5D

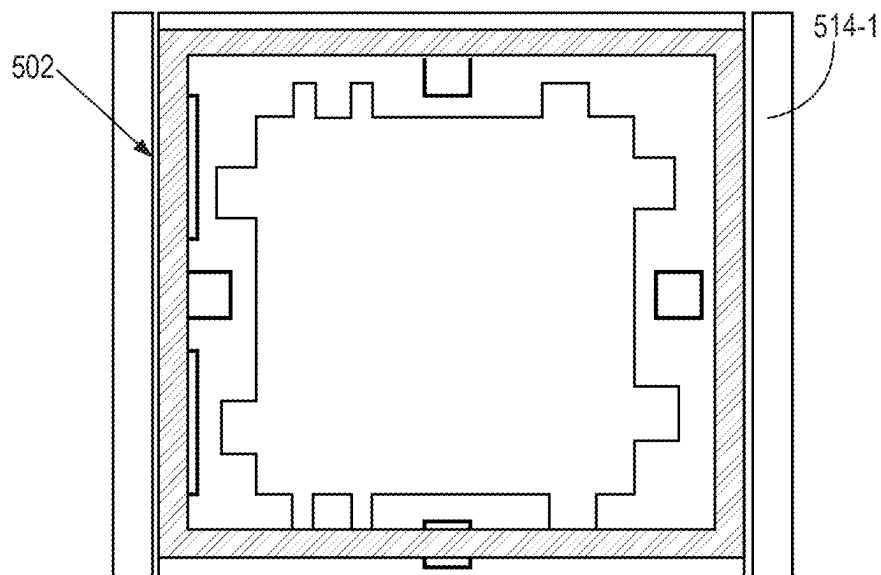


FIG. 5E

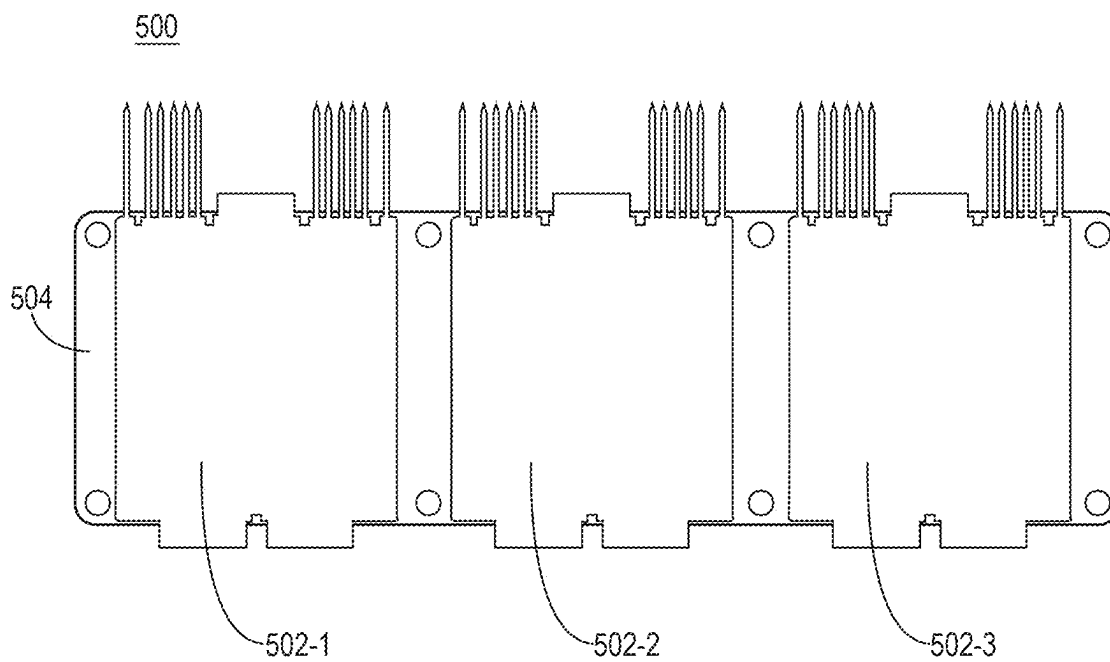


FIG. 5F

600

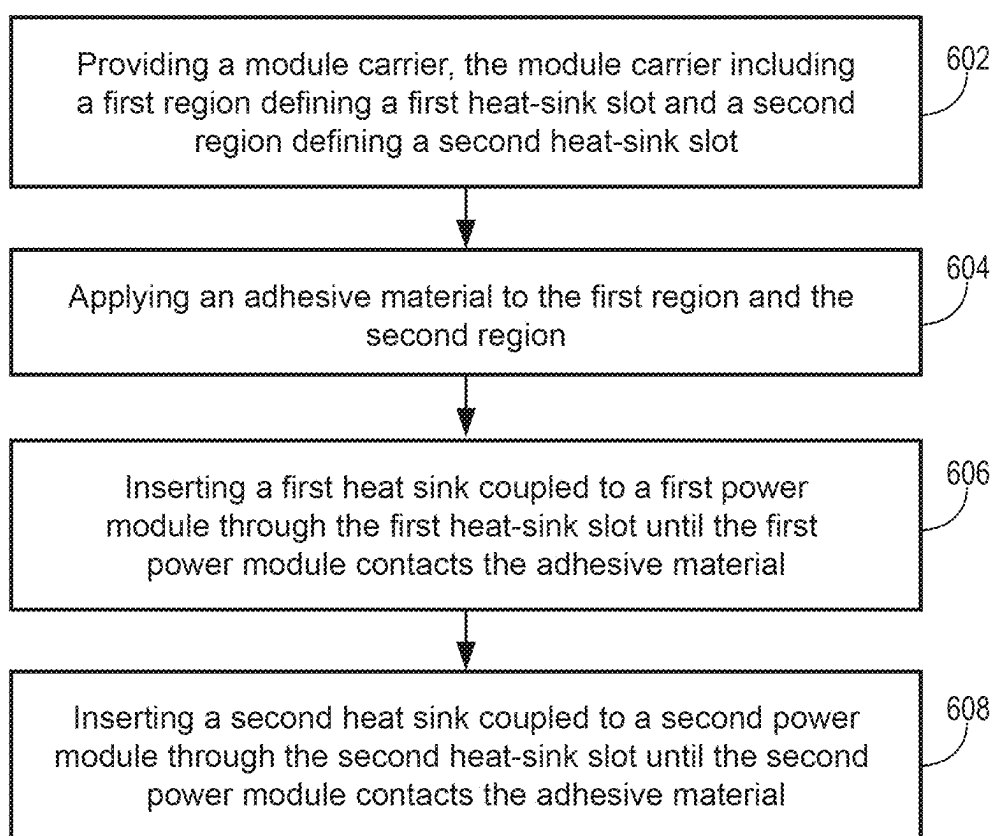


FIG. 6

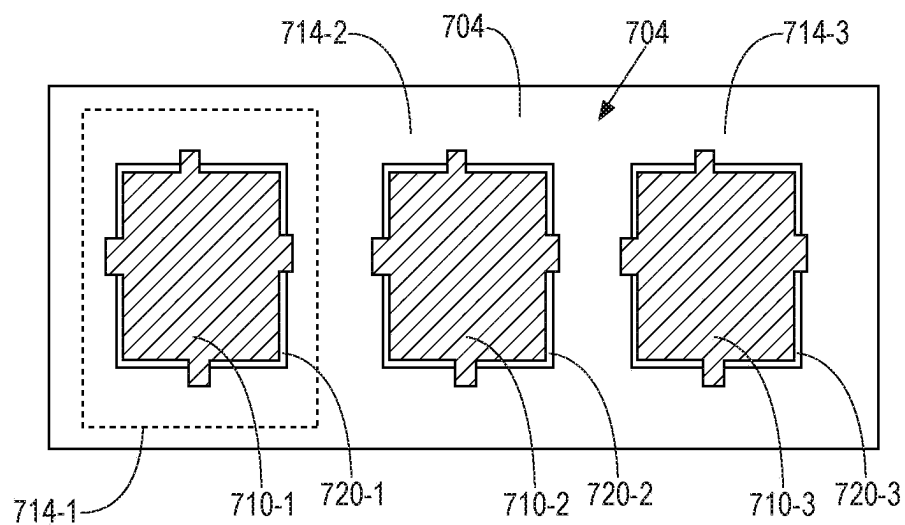


FIG. 7A

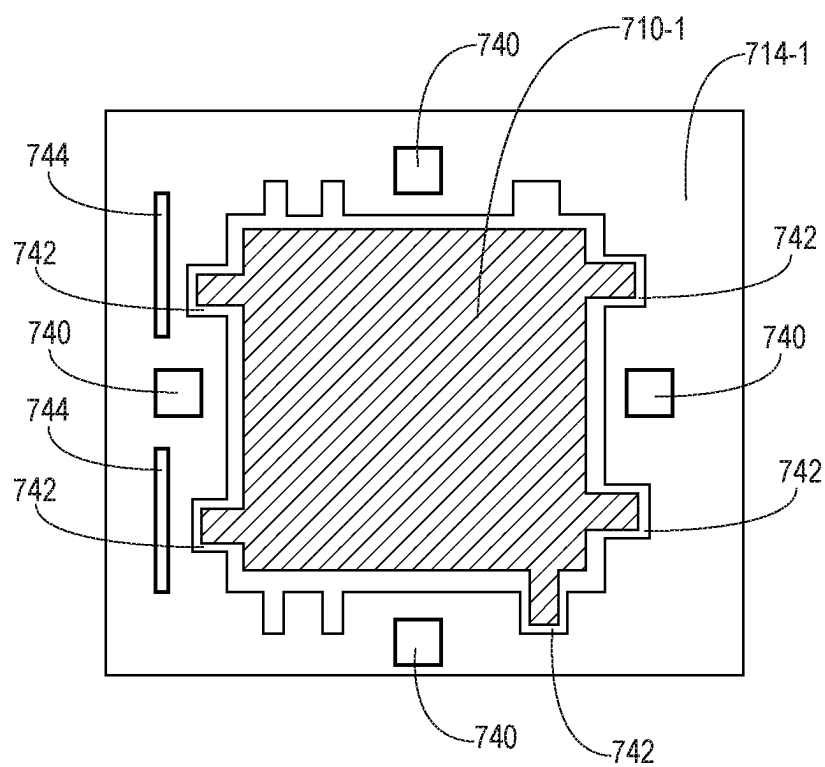


FIG. 7B

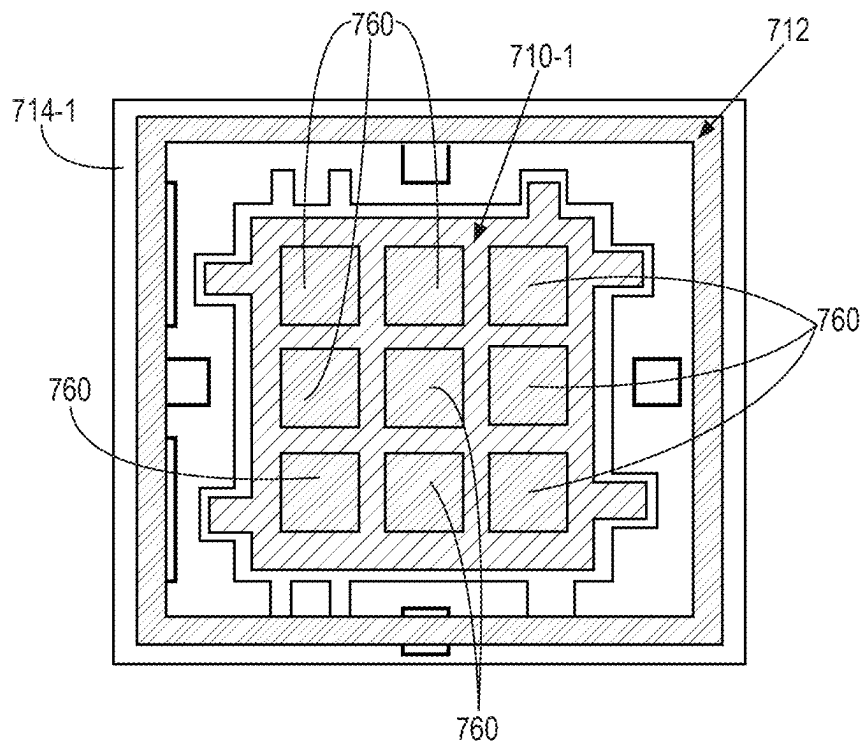


FIG. 7C

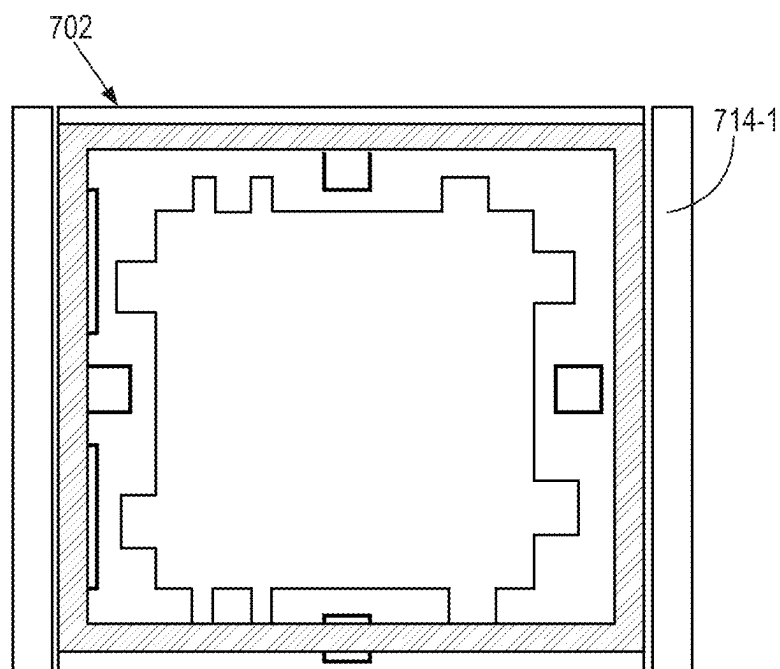


FIG. 7D

800

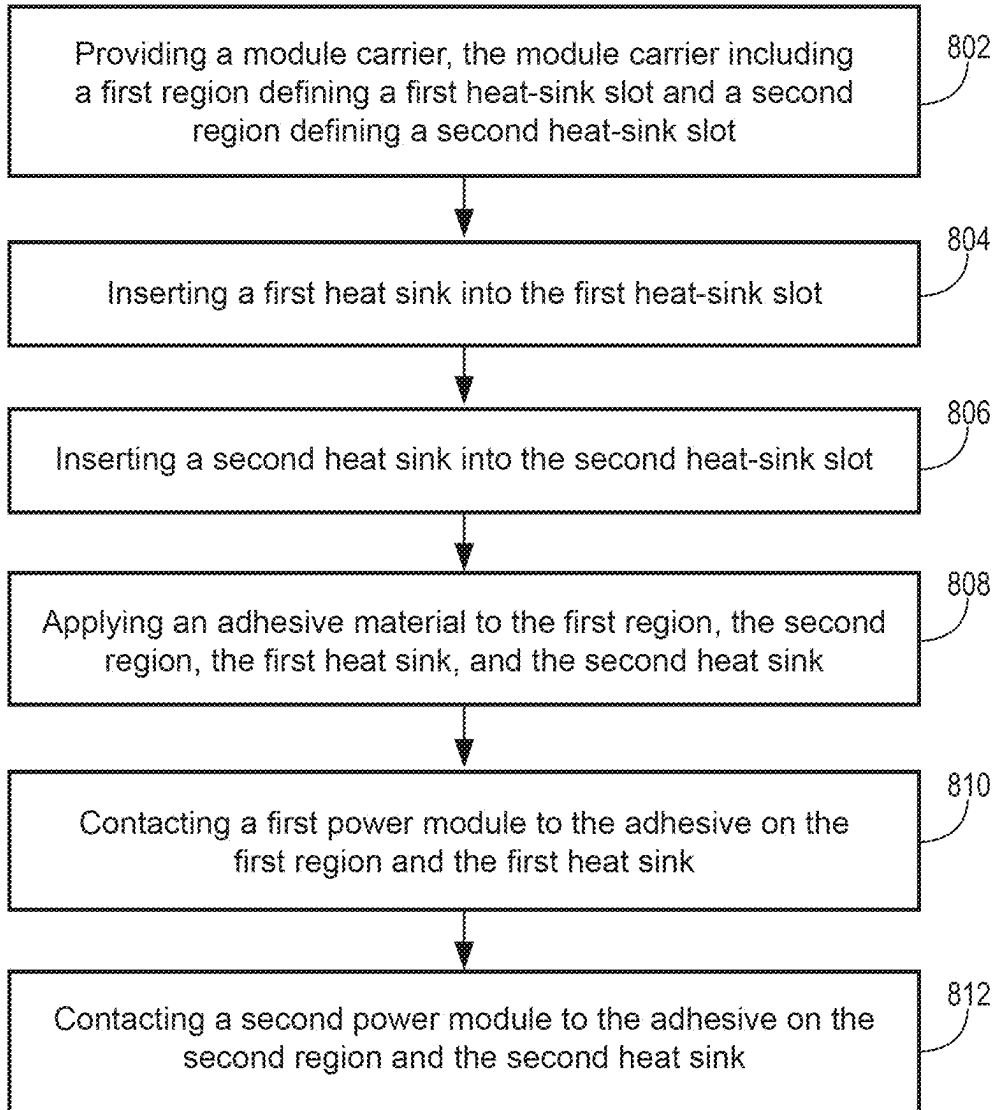


FIG. 8

POWER MODULE PACKAGE FOR DIRECT COOLING MULTIPLE POWER MODULES

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/542,230, filed Dec. 15, 2023, which is a continuation application of U.S. application Ser. No. 17/822,844, filed Aug. 29, 2022, now U.S. Pat. No. 11,848,320, which is a divisional application of U.S. patent application Ser. No. 16/949,896, filed Nov. 19, 2020, all of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] This description relates to a power module package for direct cooling multiple power modules.

BACKGROUND

[0003] A power module package may contain multiple power modules. However, some conventional power module packages may be relatively large to meet cooling performance requirements.

SUMMARY

[0004] According to an aspect, a power module package for direct cooling multiple power modules includes a plurality of power modules including a first power module and a second power module, a plurality of heat sinks including a first heat sink coupled to the first power module and a second heat sink coupled to the second power module, and a module carrier coupled to the plurality of power modules, where the module carrier includes a first region defining a first heat-sink slot and a second region defining a second heat-sink slot. The first heat sink extends at least partially through the first heat-sink slot and the second heat sink extends at least partially through the second heat-sink slot. The power module package includes a housing coupled to the module carrier, where the module carrier is disposed between the housing and the plurality of power modules. The power module package includes a ring member located between the module carrier and the housing.

[0005] According to some aspects, the power module package may include one or more of the following features (or any combination thereof). The first heat sink includes metal pin fins. The first region may include at least one positioning guide configured to align at least one of the first power module or the first heat sink. The module carrier includes a plastic material. The plurality of power modules may include a third power module and the plurality of heat sinks may include a third heat sink coupled to the third power module. The module carrier includes a third region defining a third heat-sink slot, and the third heat sink extends at least partially through the third heat-sink slot. Each of the first power module, the second power module, and the third power module may correspond to a separate phase of a three phase current of an inverter. The ring member includes an O-ring that extends around the first power module, the second power module, and the third power module, collectively. The first power module includes a half-bridge module. The module carrier is coupled to the housing via one or more fasteners.

[0006] According to an aspect, a power module package for direct cooling multiple power modules includes a plurality of power modules including a first power module, a

second power module, and a third power module, a plurality of heat sinks including a first heat sink coupled to the first power module, a second heat sink coupled to the second power module, and a third heat sink coupled to the third power module, and a module carrier coupled to the plurality of power modules, the module carrier including a first region defining a first heat-sink slot, a second region defining a second heat-sink slot, and a third region defining a third heat-sink slot. The first heat sink extends at least partially through the first heat-sink slot. The second heat sink extends at least partially through the second heat-sink slot. The third heat sink extends at least partially through the third heat-sink slot. The power package module includes a housing coupled to the module carrier, where the housing is coupled to the module carrier being located between the housing and the plurality of power modules.

[0007] According to some aspects, the power module package may include one or more of the following features (or any combination thereof). The power module package may include a ring member located between the module carrier and a surface of the housing. The first region defines one or more module positioning guides. The first region defines one or more heat-sink positioning guides. Each of the first power module, the second power module, and the third power module corresponds to a separate phase of a three phase current of an inverter. Each of the first heat sink, the second heat sink, and the third heat sink include metal pin fins. The module carrier includes a metal-based material.

[0008] According to an aspect, a method of assembly a power module package integrating multiple power modules includes receiving a module carrier, the module carrier including a first region defining a first heat-sink slot and a second region defining a second heat-sink slot, applying an adhesive material to the first region and the second region, inserting a first heat sink coupled to a first power module through the first heat-sink slot until the first power module contacts the adhesive material, and inserting a second heat sink coupled to a second power module through the second heat-sink slot until the second power module contacts the adhesive material. In some examples, each of the first region and the second region defines one or more heat-sink positioning guides.

[0009] According to an aspect, a method of assembly a power module package integrating multiple power modules includes receiving a module carrier, the module carrier including a first region defining a first heat-sink slot and a second region defining a second heat-sink slot, inserting a first heat sink into the first heat-sink slot, inserting a second heat sink into the second heat-sink slot, applying an adhesive to the first region, the second region, the first heat sink, and the second heat sink, contacting a first power module to the adhesive on the first region and the first heat sink, and contacting a second power module to the adhesive on the second region and the second heat sink. Each of the first region and the second region defines one or more module positioning guides and one or more heat-sink positioning guides.

[0010] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

- [0011] FIG. 1A illustrates a cross sectional view of a power module package according to an aspect.
- [0012] FIG. 1B illustrates a module carrier of the power module package according to an aspect.
- [0013] FIG. 1C illustrates a top view perspective of the power module package according to an aspect.
- [0014] FIGS. 2A through 2F illustrate a power module package according to an aspect.
- [0015] FIGS. 3 and 4 illustrate examples of a heat sink according to several aspects.
- [0016] FIG. 5A through 5F illustrate a power module package that is assembled according to an aspect.
- [0017] FIG. 6 illustrates a flowchart depicting example operations for assembling a power module package according to an aspect.
- [0018] FIGS. 7A through 7D illustrate a power module package that is assembled according to another aspect.
- [0019] FIG. 8 illustrates a flowchart depicting example operations for assembling a power module package according to an aspect.

DETAILED DESCRIPTION

[0020] The present disclosure relates to a power module package for direct cooling multiple power modules. The power module packages discussed herein may be compact, robust, and/or relatively easy to assemble while providing an efficient structure to permit direct cooling. The power module package includes a plurality of power modules, a plurality of heat sinks coupled to the power modules, a module carrier coupled to the power modules, a housing coupled to the module carrier, and a ring member located (e.g., disposed) between the module carrier and the housing.

[0021] In some examples, the power modules include two power modules. In some examples, the power modules include three power modules. In some examples, the power modules include half-bridge power modules. The power module package may include a first power module, a second power module, and a third power module, where each module corresponds to a separate phase of a three-phase current (UVM) of an inverter. In some examples, the power modules include more than three power modules.

[0022] The power modules are integrated into the module carrier, and the module carrier is coupled to the housing such that the module carrier is disposed between the power modules and the module carrier. In some examples, the housing is a housing for an inverter. The ring member may be disposed within a groove on the housing, and may extend around the power modules, collectively. In some examples, the power module package includes a single ring member. The module carrier may define a number of regions, where each region defines a heat-sink slot. For example, the module carrier may include a first region defining a first heat-sink slot and a second region defining a second heat-sink slot. In some examples, each of the first region and the second region includes one or more positioning guides that properly align a power module and/or a heat sink on a respective region.

[0023] The power package module may be assembled according to a first assembly process. According to the first assembly process, in a pre-processing step, the heat sinks are coupled to the power modules. For example, the first power module is coupled to a first heat sink, and the second power

module is coupled to a second heat sink. In some examples, the heat sinks include pin fin heat sinks, where a pin fin heat sink includes a base plate with a plurality of pin fins that extend from the base plate. The base plate is coupled to a surface of a power module. According to the first assembly process, an adhesive material is dispensed on each region such as the first region and the second region. The first heat sink is inserted into the first heat-sink slot until the first power module contacts the adhesive material on the first region (where the positioning guide(s) are used to properly align the first power module on the first region), and the second heat sink is inserted into the second heat-sink slot until the second power module contacts the adhesive material on the second region (where the positioning guide(s) are used to properly align the second power module on the second region). Then, the module carrier with the power modules are cured (e.g., oven cured by heat).

[0024] The power package module may be assembled according to a second assembly process. According to the second assembly process, the heat sinks are initially not coupled to the power modules, but rather the individual heat sinks are inserted into the heat-sink slots, and the power modules are then coupled to the module carrier and the heat sinks. For example, The first heat sink is inserted into the first heat-sink slot (where the positioning guide(s) are used to properly align the first heat sink on the first region), and the second heat sink is inserted into the second heat-sink slot (where the positioning guide(s) are used to properly align the second heat sink on the second region). Then, an adhesive material is dispensed on the first region, the second region, the first heat sink, and the second heat sink. The first power module is positioned on the adhesive material on the first region and the first heat sink, and the second power module is positioned on the adhesive material on the second region and the second heat sink. Then, the module carrier with the power modules are cured (e.g., oven cured by heat).

[0025] FIGS. 1A through 1C illustrate a power module package 100 for direct cooling multiple power modules 102. FIG. 1A illustrates a cross sectional view of the power module package 100. The power module package 100 includes a plurality of power modules 102, a plurality of heat sinks 110 coupled to the power modules 102, a module carrier 104 coupled to the power modules 102, a housing 106 coupled to the module carrier 104, and a ring member 108 located (e.g., disposed) between the module carrier 104 and the housing 106.

[0026] FIG. 1B illustrates an enlarged view of the module carrier 104. The module carrier 104 is disposed between the power modules 102 and the housing 106, where the module carrier 104 defines a number of heat-sink slots (e.g., 120-1, 120-2, 120-3) that receive (and align) a separate heat sink 110. FIG. 1C illustrates a top view perspective of the power module package 100 that depicts the ring member 108 extending around the power modules 102, collectively. In some examples, the power module package 100 includes a single ring member 108. The ring member 108 may be disposed between the module carrier 104 and the housing 106 and may assist with sealing a chamber 124 of the housing 106.

[0027] The power module package 100 may be a high-voltage power package. In some examples, the power module package 100 may operate at voltages greater than one thousand volts. The amount of heat generated by the power modules 102 is reduced by injecting a cooling fluid into the

chamber 124 of the housing 106. The cooling fluid may include cooled gas such as air and/or cooled liquid such as a cooling fluid and/or cooling mixture. The power modules 102 are directly cooled by the cooling fluid. Direct cooling involves the application of a cooling fluid directly on a portion of a power module 102 (e.g., a backside of a power module 102 such as a heat sink 110 coupled (e.g., directly coupled) to the power module 102). For example, since heat sinks 110 are coupled (e.g., directly coupled) to the power module 102, the application of cooling fluid to the heat sinks 110 within the chamber 124 directly cools the power modules 102. In contrast, indirect cooling involves the application of a cooling fluid to a component that is sealed from one or more power modules. The power module package 100 may be relatively compact, robust, and relatively simple to assemble. In some examples, instead of individually sealing each power module for direct cooling, the power module package 100 uses a module carrier 104 that can seal, collectively, multiple power modules 102 in a compact fashion.

[0028] A power module 102 includes one or more silicon powered dies soldered onto one or more substrates, which can be direct bonded copper (DBC) metalized ceramics soldered onto a base plate (e.g., a copper plate). In some examples, a power module 102 includes one or more silicon-based power chips (e.g., insulated-gate bipolar transistor(s) (IGBTs), metal oxide silicon field effect transistor(s) (MOSFETs), diodes, etc.) that are soldered onto a ceramic-based substrate, which is then soldered to a base plate. In some examples, the power modules 102 include half-bridge modules (e.g., an inverter leg with two switches and their corresponding antiparallel diodes). However, the power modules 102 may include other types of module topologies such as switch, bridge rectifier, H-bridge, etc.

[0029] In some examples, the power module package 100 includes half bridge direct cooling modules that are integrated on an inverter. In some examples, the power module package 100 includes a three-phase inverter circuit (e.g., a 6-pack module or IGBT 6-pack module). In some examples, the power module package 100 is configured to be used within an automotive inverter for an electric vehicle or a hybrid vehicle. For example, an automotive inverter may accelerate or decelerate the vehicle by converting the direct current (DC) power from the batteries to alternating current (AC) at the frequency required for vehicle speed and other system controls to control the electric motor speed, drive, torque, and/or power. In some examples, the housing 106 is the housing of the automotive inverter.

[0030] The power modules 102 include a first power module 102-1 and a second power module 102-2. In some examples, the first power module 102-1 is a half-bridge power module. In some examples, the second power module 102-2 is a half-bridge power module. In some examples, the power modules 102 include a third power module 102-3. In some examples, the third power module 102-3 is a half-bridge power module. In some examples, the power module package 100 includes two power modules 102. In some examples, the power module package 100 includes three power modules 102. In some examples, the power module package 100 includes more than three power modules 102. In some examples, each of the first power module 102-1, the second power module 102-2, and the third power module 102-3 correspond to a separate phase of a three-phase current (UVM) of an inverter. In some examples, the first

power module 102-1, the second power module 102-2, and the third power module 102-3, collectively, operate as a 6-pack full bridge module.

[0031] Each of the power modules 102 include a first surface 111 and a second surface 113. The distance between the first surface 111 and the second surface 113 may define a thickness of a respective power module 102 in a direction A1. Each power module 102 is connected to a separate heat sink 110. In some examples, each power module 102 is connected to a separate heat sink 110 using an adhesive material (e.g., a polymer-based material, solder material, glue material, etc.). The first power module 102-1 is coupled to a first heat sink 110-1, the second power module 102-2 is coupled to a second heat sink 110-2, and the third power module 102-3 is coupled to a third heat sink 110-3. In further detail, the first heat sink 110-1 is coupled to the second surface 113 of the first power module 102-1, the second heat sink 110-2 is coupled to the second surface 113 of the second power module 102-2, and the third heat sink 110-3 is coupled to the second surface 113 of the third power module 102-3.

[0032] The first heat sink 110-1 extends from the second surface 113 of the first power module 102-1 in the direction A1. The second heat sink 110-2 extends from the second surface 113 of the second power module 102-2 in the direction A1. The third heat sink 110-3 extends from the second surface 113 of the third power module 102-3 in the direction A1. In some examples, the first heat sink 110-1 includes a pin fin heat sink. In some examples, the second heat sink 110-2 includes a pin fin heat sink. In some examples, the third heat sink 110-3 includes a pin fin heat sink. In some examples, a pin fin heat sink includes a base plate that is coupled to the second surface 113 of a respective power module 102, and a plurality of pin fins that extend in the direction A1.

[0033] Each power module 102 is coupled to the module carrier 104 via an adhesive material 112. In some examples, the adhesive material 112 includes a polymer-based material. In some examples, the adhesive material 112 includes a solder material. In some examples, the adhesive material 112 includes a glue material. In some examples, the adhesive material 112 includes a transient liquid phase sintering (TLPS) material. The module carrier 104 includes a first surface 115 and a second surface 117. In some examples, the second surface 117 is parallel with the first surface 115. The distance between the first surface 115 and the second surface 117 may define the thickness of the module carrier 104 in the direction A1. The first surface 115 of the module carrier 104 is coupled to the second surface 113 of each of the first power module 102-1, the second power module 102-2, and the third power module 102-3. The first surface 115 is disposed in a plane A4. The direction A1 is aligned perpendicular to the plane A4, and a direction A2 is perpendicular to the direction A1. A direction A3 is orthogonal to directions A1 and A2. As shown in FIG. 1A, the direction A3 is depicted into the page (shown as a dot). However, since FIG. 1B is a top of the module carrier 104, the direction A1 on FIG. 1B is depicted into the page (as shown as a dot).

[0034] The chamber 124 may be sealed from the power modules 102 via the module carrier 104. In other words, the sealing may not be directly on the DBC material of the power modules 102, but rather the sealing is created using the module carrier 104. The module carrier 104 is a single (integral) body defining a number of heat-sink slots, where

the number of heat-sink slots corresponds to the number of power modules 102. The module carrier 104 is disposed between the power modules 102 and the housing 106. In some examples, the module carrier 104 may enable relatively easy integration with the housing 106. In some examples, the module carrier 104 is a middle plate. In some examples, the module carrier 104 includes a plastic material. In some examples, the module carrier 104 includes a metal-based material. In some examples, the module carrier 104 includes an aluminum material. In some examples, the module carrier 104 includes a copper material. The module carrier 104 has a length that extends in the direction A2, a width that extends in the direction A3, and a thickness that extends in the direction A1. In some examples, the module carrier 104 has a rectangle shape. However, the module carrier 104 may include other types of shapes including curve and/or rounded portions.

[0035] As shown in FIG. 1B, the module carrier 104 includes a first region 114-1, a second region 114-2, and a third region 114-3. The first region 114-1 is the portion of the module carrier 104 in which the first power module 102-1 is mounted. The second region 114-2 is the portion of the module carrier 104 in which the second power module 102-2 is mounted. The third region 114-3 is the portion of the module carrier 104 in which the third power module 102-3 is mounted. In some examples, if the module carrier 104 integrates three power modules 102, the module carrier 104 may have a total of three regions, where each region has the same size. In some examples, one region may be larger than another region if a power module 102 is larger than another power module 102.

[0036] The first region 114-1 defines a heat-sink slot 120-1. The size (e.g., in the directions A2 and A3) of the heat-sink slot 120-1 is smaller than the size (e.g., in the directions A2 and A3) of the first region 114-1. The second region 114-2 defines a heat-sink slot 120-2. The size (e.g., in the directions A2 and A3) of the heat-sink slot 120-2 is smaller than the size (e.g., in the directions A2 and A3) of the second region 114-2. The third region 114-3 defines a heat-sink slot 120-3. The size (e.g., in the directions A2 and A3) of the heat-sink slot 120-3 is smaller than the size (e.g., in the directions A2 and A3) of the third region 114-3.

[0037] The module carrier 104 includes a portion 126 that separates the heat-sink slot 120-1 from the heat-sink slot 120-2. The module carrier 104 includes a portion 128 that separates the heat-sink slot 120-3 from the heat-sink slot 120-2. The module carrier includes a perimeter area portion 118 that extends around the heat-sink slot 120-1, the heat-sink slot 120-2, and the heat-sink slot 120-3, collectively. In some examples, the perimeter area portion 118 is coupled to the housing 106. The heat-sink slot 120-1 has a size that is larger (e.g., slightly larger) than a size of the first heat sink 110-1, and the heat-sink slot 120-1 has a size that is smaller than a size of the first power module 102-1. The heat-sink slot 120-2 has a size that is larger (e.g., slightly larger) than a size of the second heat sink 110-2, and the heat-sink slot 120-2 has a size that is smaller than a size of the second power module 102-2. The heat-sink slot 120-3 has a size that is larger (e.g., slightly larger) than a size of the third heat sink 110-3, and the heat-sink slot 120-3 has a size that is smaller than a size of the third power module 102-3. Each of the heat-sink slots (120-1, 120-2, 120-3) extends between (and through) the first surface 115 of the module carrier 104 and the second surface 117 of the module carrier 104.

[0038] The first region 114-1 is coupled to the first power module 102-1 such that the first heat sink 110-1 at least partially extends (or fully extends) through the heat-sink slot 120-1 and into the chamber 124 of the housing 106. The second region 114-2 is coupled to the second power module 102-2 such that the second heat sink 110-2 at least partially extends (or fully extends) through the heat-sink slot 120-2 and into the chamber 124 of the housing 106. The third region 114-3 is coupled to the power module 103-3 such that the third heat sink 110-3 at least partially extends (or fully extends) through the heat-sink slot 120-4 and into the chamber 124 of the housing 106.

[0039] As shown in FIG. 1B, the first region 114-1 may define a positioning guide 116-1 that aligns at least one of the first power module 102-1 or the first heat sink 110-1 during assembly of the power module package 100. In some examples, the positioning guide 116-1 is located on (or defined by) the first region 114-1 at a location proximate to the heat-sink slot 120-1. In some examples, the positioning guide 116-1 is a feature of the heat-sink slot 120-1. In some examples, the positioning guide 116-1 is a surface feature that interacts with a surface feature of the first power module 102-1 to cause the first power module 102-1 to be aligned with the first region 114-1 of the module carrier 104. In some examples, the positioning guide 116-1 is a surface feature that interacts with a surface feature of the first heat sink 110-1 to cause the first heat sink 110-1 to be properly aligned within the heat-sink slot 120-1. In some examples, the positioning guide 116-1 includes one or more protrusions. In some examples, the positioning guide 116-1 includes one or more grooves. In some examples, the positioning guide 116-1 includes one or more slots.

[0040] The second region 114-2 may define a positioning guide 116-2 that aligns at least one of the second power module 102-2 or the second heat sink 110-2. In some examples, the positioning guide 116-2 is located on (or defined by) the second region 114-2 at a location proximate to the heat-sink slot 120-2. In some examples, the positioning guide 116-2 is a feature of the heat-sink slot 120-2. In some examples, the positioning guide 116-2 is a surface feature that interacts with a surface feature of the second power module 102-2 to cause the second power module 102-2 to be aligned with the second region 114-2 of the module carrier 104. In some examples, the positioning guide 116-2 is a surface feature that interacts with a surface feature of the second heat sink 110-2 to cause the second heat sink 110-2 to be properly aligned within the heat-sink slot 120-2. In some examples, the positioning guide 116-2 includes one or more protrusions. In some examples, the positioning guide 116-2 includes one or more grooves. In some examples, the positioning guide 116-2 includes one or more slots.

[0041] The third region 114-3 may define a positioning guide 116-3 that aligns at least one of the third power module 103-3 or the third heat sink 110-3. In some examples, the positioning guide 116-3 is located on (or defined by) the third region 114-3 at a location proximate to the heat-sink slot 120-3. In some examples, the positioning guide 116-3 is a feature of the heat-sink slot 120-3. In some examples, the positioning guide 116-3 is a surface feature that interacts with a surface feature of the third power module 103-3 to cause the third power module 103-3 to be aligned with the third region 114-3 of the module carrier 104. In some examples, the positioning guide 116-3 is a

surface feature that interacts with a surface feature of the third heat sink 110-3 to cause the third heat sink 110-3 to be properly aligned with the heat-sink slot 120-3. In some examples, the positioning guide 116-3 includes one or more protrusions. In some examples, the positioning guide 116-3 includes one or more grooves. In some examples, the positioning guide 116-3 includes one or more slots.

[0042] The housing 106 is coupled to the module carrier 104, where the module carrier 104 is positioned between the housing 106 and the power modules 102. In some examples, the housing 106 is coupled to the module carrier 104 via one or more fasteners. In some examples, the housing 106 is coupled to the module carrier 104 by welding (e.g., laser welding). In some examples, the housing 106 includes a plastic material. The housing 106 defines a chamber 124, where cooling fluid is injected into the chamber 124 via an opening 125 and expelled from the chamber 124 via an opening 125. Each of the first heat sink 110-1, the second heat sink 110-2, and the third heat sink 110-3 includes a portion that is disposed within the chamber 124. In some examples, the housing 106 is an inverter housing. In some examples, the housing 106 is an inverter housing for a vehicle.

[0043] The power module package 100 includes a ring member 108 that is located (e.g., disposed) between the housing 106 and the module carrier 104. The ring member 108 includes a portion that contacts the housing and a portion that contacts the module carrier 104. In some examples, the housing 106 may define a groove, and the ring member 108 is positioned in the groove between the housing 106 and the module carrier 104. The ring member 108 extends around the first power module 102-1, the second power module 102-2, and the third power module 102-3, collectively. In some examples, the ring member 108 includes an O-ring. In some examples, the power module package 100 includes a single ring member 108.

[0044] FIGS. 2A through 2F illustrate a power module package 200 for direct cooling multiple power modules 202. The power module package 200 may be an example of the power module package 100 of FIGS. 1A through 1C. The power module package 200 includes a plurality of power modules 202, a plurality of heat sinks 210 coupled to the power modules 202, a module carrier 204 coupled to the power modules 202, a housing 206 coupled to the module carrier 204, and a ring member 208 located (e.g., disposed) between the module carrier 204 and the housing 206. FIG. 2A illustrates a perspective of the power module package 200. FIG. 2B illustrates a perspective of the power module package 200 with the power modules 202 being transparent. FIG. 2C illustrates a top view perspective of the module carrier 204. FIG. 2D illustrates a region of the module carrier 204. FIG. 2E illustrates a cross sectional view of the power module package 200. FIG. 2F illustrates a side view of a power module 202.

[0045] The power modules 202 include a first power module 202-1, a second power module 202-2, and a third power module 202-3. In some examples, each of the first power module 202-1, the second power module 202-2, and the third power module 202-3 correspond to a separate phase of a three-phase current (UVM) of an inverter. In some examples, the first power module 202-1, the second power module 202-2, and the third power module 202-3, collectively, operate as a 6-pack full bridge module. In some examples, the power modules 202 includes terminals 230

and contacts 232. In some examples, the terminals 230 extend in the direction A1, and the contacts 232 extend in the direction A2.

[0046] As shown in FIG. 2F, a power module 202 includes a first surface 211 and a second surface 213. The distance between the first surface 211 and the second surface 213 may define a thickness of a respective power module 202 in a direction A1. Each power module 202 is connected to a separate heat sink 210. In some examples, each power module 202 is connected to a separate heat sink 210 using an adhesive material. The first power module 202-1 is coupled to a first heat sink 210-1, the second power module 202-2 is coupled to a second heat sink 210-2, and the third power module 202-3 is coupled to a third heat sink 210-3. In further detail, the first heat sink 210-1 is coupled to the second surface 213 of the first power module 202-1, the second heat sink 210-2 is coupled to the second surface 213 of the second power module 202-2, and the third heat sink 210-3 is coupled to the second surface 213 of the third power module 202-3.

[0047] The first heat sink 210-1 extends from the second surface 213 of the first power module 202-1 in the direction A1. The second heat sink 210-2 extends from the second surface 213 of the second power module 202-2 in the direction A1. The third heat sink 210-3 extends from the second surface 213 of the third power module 202-3 in the direction A1. Each of the first heat sink 210-1, the second heat sink 210-2, and the third heat sink 210-3 includes a pin fin heat sink.

[0048] Each power module 202 is coupled to the module carrier 204 via an adhesive material 212. The module carrier 204 includes a first surface 215 and a second surface 217. In some examples, the second surface 217 is parallel with the first surface 215. The distance between the first surface 215 and the second surface 217 may define the thickness of the module carrier 204 in the direction A1. The first surface 215 of the module carrier 204 is coupled to the second surface 213 of each of the first power module 202-1, the second power module 202-2, and the third power module 202-3. The first surface 215 is disposed in a plane A4.

[0049] The chamber 224 may be sealed from the power modules 202 via the module carrier 204. In other words, the sealing may not be directly on the DBC material of the power modules 102. The module carrier 204 is a single (integral) body defining a number of heat-sink slots, where the number of heat-sink slots corresponds to the number of power modules 202. The module carrier 204 is disposed between the power modules 202 and the housing 206. In some examples, the module carrier 204 is a middle plate. In some examples, the module carrier 204 includes a plastic material. In some examples, the module carrier 204 includes a metal-based material. In some examples, the module carrier 204 includes an aluminum material. In some examples, the module carrier 204 includes a copper material. The module carrier 204 has a length that extends in the direction A2, a width that extends in the direction A3, and a thickness that extends in the direction A1. In some examples, the module carrier 204 has a rectangle shape. However, the module carrier 204 may include other types of shapes includes curved or rounded portions.

[0050] As shown in FIG. 2C, the module carrier 104 includes a first region 214-1, a second region 214-2, and a third region 214-3. The first region 214-1 defines a heat-sink slot 220-1. The second region 214-2 defines a heat-sink slot

220-2. The third region **214-3** defines a heat-sink slot **220-3**. The module carrier **204** includes a portion **226** that separates the heat-sink slot **220-1** from the heat-sink slot **220-2**. The module carrier **204** includes a portion **228** that separates the heat-sink slot **220-3** from the heat-sink slot **220-2**. The module carrier **204** includes a perimeter area portion **218** that extends around the heat-sink slot **220-1**, the heat-sink slot **220-2**, and the heat-sink slot **220-3**, collectively. In some examples, the module carrier **204** defines openings **236** that are used to fasten the module carrier **204** to the housing **206**. In some examples, the openings **236** are defined on the perimeter area portion **218** at the corner portions of the module carrier **204**. As shown in FIG. 2E, fasteners **205** may extend through the openings **236** and into the housing **206**. In some examples, fasteners **205** are not used, but rather the module carrier **204** is coupled to the housing **206** via other coupling mechanisms such as welding (e.g., laser welding).

[0051] The heat-sink slot **220-1** has a size that is larger than a size of the first heat sink **210-1**, and the heat-sink slot **220-1** has a size that is smaller than a size of the first power module **202-1**. The heat-sink slot **220-2** has a size that is larger than a size of the second heat sink **210-2**, and the heat-sink slot **220-2** has a size that is smaller than a size of the second power module **202-2**. The heat-sink slot **220-3** has a size that is larger than a size of the third heat sink **210-3**, and the heat-sink slot **220-3** has a size that is smaller than a size of the third power module **202-3**. Each of the heat-sink slots (**220-1**, **220-2**, **220-3**) extends between (and through) the first surface **215** of the module carrier **204** and the second surface **217** of the module carrier **204**.

[0052] The first region **214-1** is coupled to the first power module **202-1** such that the first heat sink **210-1** extends through the heat-sink slot **220-1** and into the chamber **224** of the housing **206**. The second region **214-2** is coupled to the second power module **202-2** such that the second heat sink **210-2** extends through the heat-sink slot **220-2** and into the chamber **224** of the housing **206**. The third region **214-3** is coupled to the power module **203-3** such that the third heat sink **210-3** extends through the heat-sink slot **220-4** and into the chamber **224** of the housing **206**.

[0053] As shown in FIG. 2D, in some examples, the first region **214-1** may define module positioning guides **240**. The module positioning guides **240** may be an example of the positioning guide **116-1** of FIG. 1B. The module positioning guides **240** are surface features configured to interact with surface features on the first power module **202-1** to properly align the first power module **202-1** with the first region **214-1**. Although four module positioning guides **240** are depicted in FIG. 2D, the first region **214-1** may define any number of module positioning guides **240**. In some examples, the module positioning guides **240** include protrusions that extend from the second surface **217** of the module carrier **204**, where the protrusions interact with slots defined by the first power module **202-1**. In some examples, the module positioning guides **240** include slots that extend into the second surface **217** of the module carrier **204**, where the slots interact with protrusions defined by the first power module **202-1**.

[0054] In some examples, the first region **214-1** may include heat-sink positioning guides **242**. The heat-sink positioning guides **242** may be an example of the positioning guide **116-1** of FIG. 1B. In some examples, the first region **214-1** includes the module positioning guides **240** but not the heat-sink positioning guides **242**. In some examples, the

first region **214-1** includes the heat-sink positioning guides **242** but not the module positioning guides **242**. In some examples, the first region **214-1** includes the module positioning guides **240** and the heat-sink positioning guides **242**.

[0055] The heat-sink positioning guides **242** are surface features configured to interact with surface features on the first heat sink **210-1** to properly align the first heat sink **210-1** within the heat-sink slot **220-1**. Although six heat-sink positioning guides **242** are depicted in FIG. 2D, the first region **214-1** may define any number of heat-sink positioning guides **242**. In some examples, the heat-sink positioning guides **242** include slots that extend from the first heat-sink slot **220-1**. The second region **214-2** and the third region **214-3** may include the same features as discussed with reference to the first region **214-1**. In some examples, the first region **214-1** includes one or more adhesive stop features **244**. In some examples, the adhesive stop features **244** are slots that extend into the first region **214-1** (e.g., having a certain depth). The adhesive stop features **244** may accept adhesive overflow during the assembly process.

[0056] The housing **206** is coupled to the module carrier **204**, where the module carrier **204** is positioned between the housing **206** and the power modules **202**. In some examples, the housing **206** is coupled to the module carrier **204** via one or more fasteners **205**. In some examples, the housing **206** is coupled to the module carrier **204** by welding (e.g., laser welding). In some examples, the housing **206** includes a plastic material. The housing **206** defines a chamber **224**, where cooling fluid is injected into the chamber **224** via an opening **225** and expelled from the chamber **224** via an opening **225**. Each of the first heat sink **210-1**, the second heat sink **210-2**, and the third heat sink **210-3** includes a portion that is disposed within the chamber **224**. In some examples, the housing **206** is an inverter housing. In some examples, the housing **206** is an inverter housing for a vehicle.

[0057] The power module package **200** includes a ring member **208** that is located (e.g., disposed) between the housing **206** and the module carrier **204**. The ring member **208** includes a portion that contacts the housing and a portion that contacts the module carrier **204**. In some examples, the housing **206** may define a groove **271**, and the ring member **208** is positioned in the groove **271** between the housing **206** and the module carrier **204**. The ring member **208** extends around the first power module **202-1**, the second power module **202-2**, and the third power module **202-3**, collectively. In some examples, the ring member **208** includes an O-ring. In some examples, the power module package **200** includes a single ring member **208**.

[0058] FIG. 3 illustrates an example of a heat sink **310**. The heat sink **310** may be any of the heat sinks discussed with reference to the previous figures. The heat sink **310** is a pin fin heat sink. For example, the heat sink **310** includes a base plate **323** and a plurality of pin fins **321** that extend from the base plate **323**. In some examples, as shown in FIG. 3, the pin fins **321** have the same length. FIG. 4 illustrates an example of a heat sink **410**. The heat sink **410** may be any of the heat sinks discussed with reference to the previous figures. The heat sink **410** is a pin fin heat sink. For example, the heat sink **410** includes a base plate **423** and a plurality of pin fins **421** that extend from the base plate **423**. In some examples, as shown in FIG. 4, the pin fins **421** have different lengths.

[0059] FIGS. 5A through 5F illustrate a power module package 500 that is assembled according to a first assembly process. The power module package 500 may be an example of any of the power module packages discussed with reference to the previous figures. FIG. 5A illustrates a power module 502 with an attached heat sink 510, which may be an example of a first power module 502-1, a second power module 502-2, and a third power module 502-3. Also, the power module 502 includes terminals 530 and contacts 532. FIG. 5B illustrates a module carrier 504 defining a first region 514-1, a second region 514-2, and a third region 514-3. The first region 514-1 defines a first heat-sink slot 520-1. The second region 514-2 defines a second heat-sink slot 520-2. The third region 514-3 defines a third heat-sink slot 520-3. Also, the module carrier 504 may include a plurality of openings 536. FIG. 5C illustrates a perspective of the first region 514-1 that defines module positioning guides 540, heat-sink positioning guides 542, and/or adhesive stop features 544. FIG. 5D illustrates an adhesive material 512 applied to the first region 514-1, where the adhesive stop features 544 accept adhesive overflow from the application of the adhesive material 512 to the first region 514-1. FIG. 5E illustrates the power module 502 coupled to the first region 514-1. FIG. 5F illustrates a first power module 502-1, a second power module 502-2, and a third power module 502-3 coupled to the module carrier 504 after the adhesive material 512 is cured.

[0060] Before the first assembly process starts, the heat sinks 510 are already coupled to the power modules 502. With respect to the first region 514-1, an adhesive material 512 is applied to the first region 514-1. Then, the heat sink 510 that is coupled to the first power module 502-1 is inserted into the first heat-sink slot 520-1, where the heat-sink positioning guides 542 guide the heat sink 510 through the heat-sink slot 520-1 such that the first power module 502-1 contacts the adhesive material 512. The same process happens with respect to the second region 514-2 and the third region 514-3. For example, the adhesive material 512 is applied to the first region 514-1, the second region 514-2, and the third region 514-3. Then, the first power module 502-1, the second power module 502-2, and the third power module 502-3 are positioned on the module carrier 504 in the same manner as discussed with reference to the first region 514-1. Then, the module carrier 504 with the first power module 502-1, the second power module 502-2, and the third power module 502-3 may be cured (e.g., oven cured) to cure the adhesive material 512.

[0061] FIG. 6 depicts a flowchart 600 depicting example assembly operations for assembling a power module package according to an aspect. Although the flowchart 600 is described with reference to the power module package 100, the flowchart may be applicable to any of the power module packages discussed herein. Although the flowchart 600 of FIG. 6 illustrates operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further, operations of FIG. 6 and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion.

[0062] Operation 602 includes providing a module carrier 104, where the module carrier 104 includes a first region 114-1 defining a first heat-sink slot 120-1 and a second region 114-2 defining a second heat-sink slot 120-2. Operation 604 includes applying an adhesive material 112 to the

first region 114-1 and the second region 114-2. Operation 606 includes inserting a first heat sink 110-1 coupled to a first power module 102-1 through the heat-sink slot 120-1 until the first power module 102-1 contacts the adhesive material 112. Operation 608 includes inserting a second heat sink 110-2 coupled to a second power module 102-2 through the heat-sink slot 120-2 until the second power module 102-2 contacts the adhesive material 112.

[0063] FIGS. 7A through 7D illustrate a power module package that is assembled according to a second assembly process. The power module package of FIGS. 7A through 7D may be an example of any of the power module packages discussed with reference to the previous figures. FIG. 7A illustrates a module carrier 704 defining a first region 714-1, a second region 714-2, and a third region 714-3. The first region 714-1 defines a first heat-sink slot 720-1. The second region 714-2 defines a second heat-sink slot 720-2. The third region 714-3 defines a third heat-sink slot 720-3. In contrast to FIGS. 5A through 5F, the heat sinks (e.g., a first heat sink 710-1, a second heat sink 710-2, a third heat sink 710-3) are not coupled to the power modules at the start of the second assembly process. Rather, the first heat sink 710-1 is inserted into the first heat-sink slot 720-1, the second heat sink 710-2 is inserted into the second heat-sink slot 720-2, and the third heat sink 710-3 is inserted into the third heat-sink slot 720-3. FIG. 7B illustrates a perspective of the first region 714-1 that defines module positioning guides 740, heat-sink positioning guides 742, and/or adhesive stop features 744. FIG. 7C illustrates an adhesive material 712 applied to the first region 714-1 and the first heat sink 710-1. In some examples, separate (small) portions 760 are applied to the first heat sink 710-1. In some examples, a single large portion 760 is applied to the first heat sink 710-1. FIG. 7E illustrates a power module 702 coupled to the first region 714-1 and the first heat sink 710-1 via the adhesive material 712.

[0064] Before the second assembly process starts, the first heat sink 710-1, the second heat sink 710-2, the third heat sink 710-3 are not coupled to the power modules. Rather, the first heat sink 710-1, the second heat sink 710-2, the third heat sink 710-3 are inserted into the first heat-sink slot 720-1, the second heat-sink slot 720-2, and the third heat-sink slot 720-3, respectively. Then, the adhesive material 712 is applied to each of the first region 714-1, the second region 714-2, and the third region 714-3, the first heat sink 710-1, the second heat sink 710-2, and the third heat sink 710-3. Then, a first power module is attached to the first region 714-1 and the first heat sink 710-1, a second power module is attached to the second region 714-2 and the second heat sink 710-2, and a third power module is attached to the third region 714-3 and the third heat sink 710-3. Then, the module carrier 704 with the first power module, the second power module, and the third power module may be cured (e.g., oven cured) to cure the adhesive material 712.

[0065] FIG. 8 depicts a flowchart 800 depicting example assembly operations for assembling a power module package according to an aspect. Although the flowchart 800 is described with reference to the power module package 100, the flowchart may be applicable to any of the power module packages discussed herein. Although the flowchart 800 of FIG. 8 illustrates operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further,

operations of FIG. 8 and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion.

[0066] Operation 802 includes providing a module carrier 104, where the module carrier 104 includes a first region 114-1 defining a first heat-sink slot 120-1 and a second region 114-2 defining a second heat-sink slot 120-2. Operation 804 includes inserting a first heat sink 110-1 into the first heat-sink slot 120-1. Operation 806 includes inserting a second heat sink 110-2 into the second heat-sink slot 120-2. Operation 808 includes applying an adhesive material 112 to the first region 114-1, the second region 114-2, the first heat sink 110-1, and the second heat sink 110-2. Operation 810 includes contacting a first power module 102-1 to the adhesive material 112 on the first region 114-1 and the first heat sink 110-1. Operation 812 includes contacting a second power module 102-2 to the adhesive material 112 on the second region 114-2 and the second heat sink 110-2.

[0067] It will be understood that, in the foregoing description, when an element is referred to as being connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly connected to or directly coupled to another element, there are no intervening elements. Although the terms directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly connected or directly coupled can be referred to as such. The claims of the application, if any, may be amended to recite exemplary relationships described in the specification or shown in the figures. Implementations of the various techniques described herein may be implemented in (e.g., included in) digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. Portions of methods also may be performed by, and an apparatus may be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

[0068] Some implementations may be implemented using various semiconductor processing and/or packaging techniques. Some implementations may be implemented using various types of semiconductor processing techniques associated with semiconductor substrates including, but not limited to, for example, Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), Silicon Carbide (SiC) and/or so forth.

[0069] While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the embodiments. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The embodiments described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different embodiments described.

What is claimed is:

1. A power module package comprising:
 - a plurality of power modules;
 - a plurality of heat sinks coupled to the plurality of power modules;
 - a module carrier coupled to the plurality of power modules, the module carrier including a plurality of regions, a region of the plurality of regions defining a heat-sink slot, a heat sink of the plurality of heat sinks extending at least partially through the heat-sink slot; and
 - a housing coupled to the module carrier.
2. The power module package of claim 1, wherein the heat sink includes a plurality of metal pin fins.
3. The power module package of claim 1, wherein the region includes at least one positioning guide configured to align a power module of the plurality of power modules.
4. The power module package of claim 1, wherein the module carrier includes a polymer material.
5. The power module package of claim 1, wherein the region is a first region, wherein the plurality of power modules include a first power module, a second power module, and a third power module, wherein the plurality of regions include a second region and a third region, wherein the first power module, the second power module, and the third power module are coupled to the first region, the second region, and the third region, respectively.
6. The power module package of claim 5, wherein each of the first power module, the second power module, and the third power module correspond to a separate phase of a three phase current of an inverter.
7. The power module package of claim 1, wherein the housing includes a chamber configured to receive cooling fluid.
8. The power module package of claim 1, wherein the module carrier is disposed between the housing and the plurality of power modules.
9. The power module package of claim 1, wherein the module carrier is coupled to the housing via one or more fasteners.
10. The power module package of claim 1, further comprising:
 - a ring member located between the module carrier and the housing.
11. The power module package of claim 10, wherein the ring member includes a closed loop that extends around the plurality of power modules, collectively.
12. A power module package comprising:
 - a plurality of power modules;
 - a plurality of heat sinks coupled to the plurality of power modules;
 - a module carrier coupled to the plurality of power modules, the module carrier including a plurality of regions, a region of the plurality of regions defining a heat-sink slot, a heat sink of the plurality of heat sinks extending at least partially through the heat-sink slot; and
 - a housing coupled to the module carrier, the housing including a chamber configured to receive cooling fluid, the heat sink having a portion that extends into the chamber.
13. The power module package of claim 12, wherein the module carrier is located between the housing and the plurality of power modules.
14. The power module package of claim 12, further comprising:

a ring member located between the module carrier and a surface of the housing, wherein the ring member includes a closed loop that extends around the plurality of power modules, collectively.

15. The power module package of claim **12**, wherein the region includes one or more module positioning guides, wherein the region includes one or more heat-sink positioning guides.

16. The power module package of claim **12**, wherein the module carrier includes a metal-based material.

17. A power module package comprising:

a plurality of power modules;

a plurality of heat sinks coupled to the plurality of power modules;

a module carrier coupled to the plurality of power modules, the module carrier including a plurality of heat-sink slots, a heat sink of the plurality of heat sinks

extending at least partially through a heat-sink slot of the plurality of heat-sink slots;

a housing coupled to the module carrier; and

a ring member located between the module carrier and the housing.

18. The power module package of claim **17**, wherein the housing includes a chamber configured to receive cooling fluid, the heat sink having a portion that extends into the chamber.

19. The power module package of claim **17**, wherein the module carrier includes one or more module positioning guides, wherein the module carrier includes one or more heat-sink positioning guides.

20. The power module package of claim **17**, wherein the plurality of power modules includes a first power module, a second power module, and a third power module.

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